TCP/IP SECURITY LAYER PROTOCOLS

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Premium Academic Assignment

1. Introduction

TCP/IP is the fundamental communication model used in modern networks. However, it does not inherently provide strong security. Therefore, multiple security protocols are applied at different layers to ensure safe communication.

2. Objectives of TCP/IP Security

The main objectives include confidentiality, integrity, authentication, and availability. These objectives protect data from unauthorized access and cyber threats.

3. Security Protocols by Layer

3.1 Application Layer

HTTPS, PGP, S/MIME secure user-level communication.

3.2 Transport Layer

SSL/TLS provide encrypted communication and authentication.

3.3 Internet Layer

IPSec secures IP packets using AH and ESP.

3.4 Network Access Layer

WPA2/WPA3 secure wireless networks.

4. Summary Table

Layer	Protocols	Function
Application	HTTPS, PGP	User security
Transport	TLS/SSL	Encryption
Internet	IPSec	Packet security
Access	WPA2/WPA3	Network protection

5. Common Network Attacks

Man-in-the-middle attacks, spoofing, sniffing, and replay attacks are common threats that TCP/IP security protocols aim to prevent.

6. Advantages of TCP/IP Security

- Provides layered protection - Enhances data privacy - Ensures reliable communication

7. Conclusion

TCP/IP security protocols ensure safe and reliable data communication. Their layered approach makes networks resilient to various cyber threats.

8. References

William Stallings, Cryptography and Network Security (5th Edition).